Panchip Microelectronics Co., Ltd.

PAN3029/3060 Series CAD Application Reference Documentation

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documentation

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revision history (of a document, web page etc)

releases	revision time	descriptive
V1.0	2023.06	initial version
V1.1	2023.08	Modifying CAD Pins
V1.2	2024.03	Modifying CAD Function Interface Formal Parameters
V1.3	2024.04	Modify the function interface and chip description

catalogs

1 Functions

PAN3029/3060 series chips support CAD-IRQ interrupt. When the CAD function is enabled and enters the Rx mode, the chip will detect whether there is a ChirpIOT™ signal in the channel, and if there is, it will set the CAD-IRQ to high, so that an external MCU can determine whether there is a ChirpIOT™ signal in the channel by detecting whether the CAD-IRQ signal is high or not within a certain period of time. The external MCU can determine whether there is a ChirpIOT™ signal in the channel by detecting whether the CAD-IRQ signal is pulled high within a certain period of time.

The user can read the CAD-IRQ signal through the GPIO port, and the channel active detection flow is shown Figure 1-1:

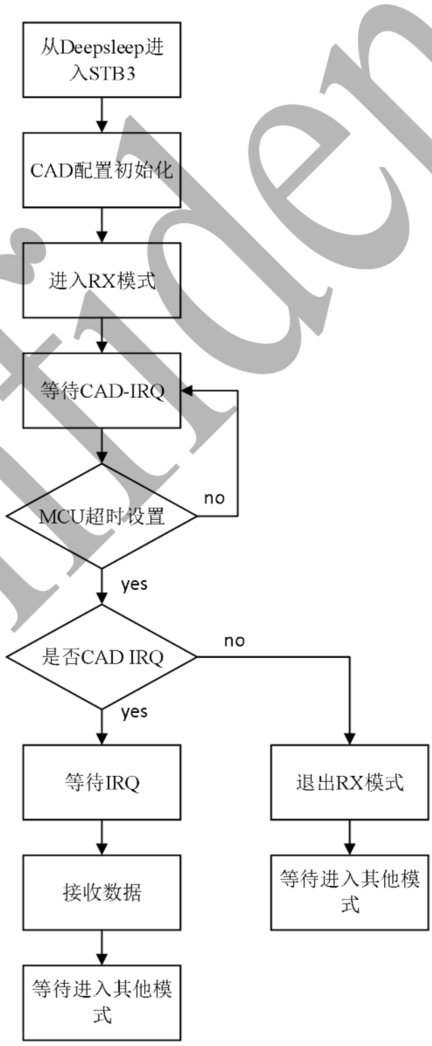

Figure 1-1 Channel Active Detection (CAD)

2 Software Design Reference

2.1 Software Design Process

1. Chip initialization;
2. Configure CAD initialization;
3. The chip enters the receive mode;
4. Observe the CAD-IRQ signal.

2.2 Software Design Verification

2.2.1 verification step

1. The sending module sends out data packets periodically;
2. The receive module is configured for receive mode;
3. Use an oscilloscope to capture the CAD-IRQ signal at the receiving end.

2.2.2 SDK Example

Reference Code:

```
ret= rf_init(); if(ret ! =
OK)
{
    printf("    RF Init Fail");
    while(1);
}
rf_set_default_para().
rf_set_cad(CAD_DETECT_THRESHOLD_10,  CAD_DETECT_NUMBER_3);// enable
CAD mode and set the Chirp-IoT signal threshold and the number of preamble valid signals
rf_enter_continous_rx().
```

```
while (1);//wait for the oscilloscope to detect the CAD-IRQ signal
```

The sample code configures CAD initialization, configures GPIO11 as the CAD detection IO, and then enters the receive mode.

The transmitter module periodically sends out data packets (the duration of the packet preamble+payload is about 42ms), capture the GPIO11 waveform of the receiver module with a logic analyzer, and observe the logic analyzer results.

2.2.3 Verification results

The oscilloscope capture results are shown in Figure 2-1:

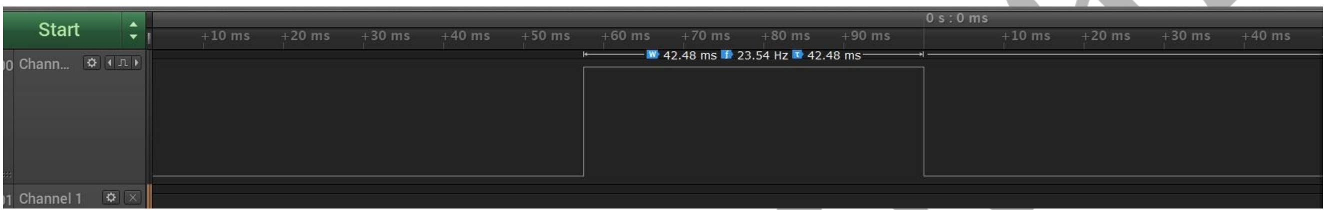

Figure 2-1 Oscilloscope Capture Results (CAD)

According to the results, CAD-IRQ occurs in the receiving module when the transmitting module sends a data packet, CAD detects pin GPIO11 is pulled up about 42ms, maintaining the length of time for a full ChirpIOT™ packet.

3 caveat

3.1 About CAD Affects the Receive Sensitivity of the Chip

When the CAD function is initialized, the chip's receive signal detection threshold and the threshold for the number of valid signals for CAD are modified. Setting different receive thresholds and the number of valid signals affects the chip's receive sensitivity and may result in CAD false triggering.

```
uint32_t rf_cad_on(uint8_t threshold, uint8_t chirps)
{
    rf_set_gpio_output(11).
```

```
        RF_ASSERT(rf_reset_spec_page_reg_bits(PAGE0_SEL, 0x5E, BIT6));
        RF_ASSERT(rf_write_spec_page_reg_bits(PAGE1_SEL, 0x25, chirps, BIT0|BIT1) );
        RF_ASSERT(rf_write_spec_page_reg(PAGE1_SEL, 0x0f, threshold)).
        return OK;
    }
```

The setting of receiving threshold needs to modify the passing parameter of rf_cad_on(uint8_t threshold, uint8_t chirps) function, and the effect of modifying receiving threshold on receiving sensitivity and false trigger probability is as follows (experimental data is tested in daily application scenario in office):

Table 3-1 Receive Threshold Settings

reception threshold	Number of active signals	false trigger probability	receiver sensitivity
0x0a	0x03	Lower(once every few hours)	not affect
0x10	0x03	Very low(0 times in 24 hours)	Deterioration 1dBm

Users need to choose to modify the passed parameters in the rf_cad_on(uint8_t threshold, uint8_t chirps) function according to the application scenario when using the CAD function. After using the CAD function, it is recommended to call the rf_cad_off() function, which can turn off the CAD function and set the receive threshold back.

3.2 About the SDK and Demo Board

The interface functions required for the CAD function are provided in the SDK. When CAD-IRQ is triggered, the detect pin GPIO11 is pulled high.

GPIO11 is connected to PA09 of the demo system board MCU.

```
#define GPIO_PIN_CAD                Pin09
#define GPIO_PORT_CAD    PortA
#define        CHECK_CAD()          PORT_GetBit(GPIO_PORT_CAD, GPIO_PIN_CAD);
```

3.3 About CAD Usage

PAN3029/3060 enables CAD detection of preamble and payload.

3.3.1 The preamble checking method

When the full preamble+payload signal arrives, the user can read the CAD-IRQ on the receiver side via the GPIO port signal, the CAD detection pin GPIO11 is pulled high for the duration preamble+payload. At this time, the

The end can produce correct rxdone results.

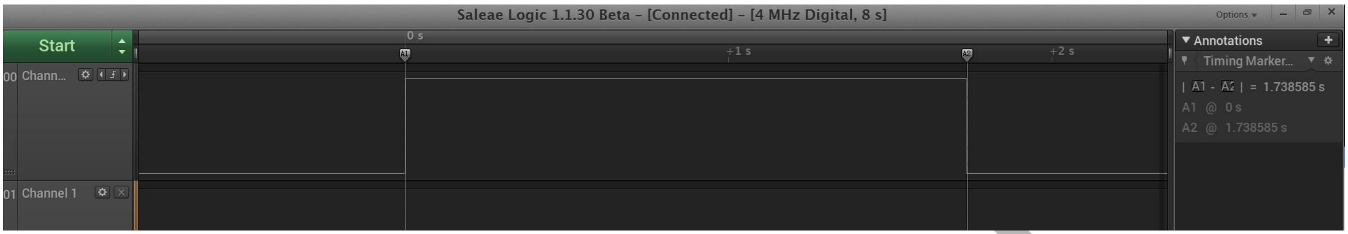

Figure 3-1 Logic Capture Results (complete preamble)

When an incomplete preamble+payload signal arrives (the transmitter first transmits the data, and then the receiver turns on CAD detection during the preamble time period), at this time, the user can read the CAD-IRQ signal through the GPIO port on the receiver side, and the change of the CAD detection pin GPIO11 can be in two cases:

1. If the preamble is more complete and the number of internally contained preamble is greater than the configured chirp_num, GPIO11 will be pulled up for the duration of preamble+payload (the preamble is not complete, so the overall GPIO11 pull-up time is shorter than the case in Figure 3-1). At this point, the receiver can produce the correct rxdone result.

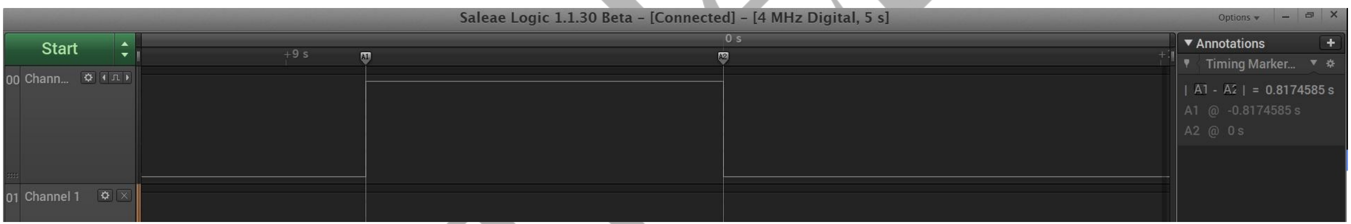

Figure 3-2 Logic Capture Results (more complete preamble)

2、If the preamble is incomplete and the number of internal preamble is less than the configured chirp_num, GPIO11 will remain low. At this time, the receiver will not generate correct reception results.

3、When there is only payload signal, GPIO11 will be low all the time. In this case, the receiver will not generate correct reception results.

3.3.2 Software Application Reference

The time for CAD detection is calculated in units of single chirp duration, which is 2^SF/BW (SF).

(is the spreading factor, BW is the bandwidth, BW is in Hz, and time is in

seconds). A software design reference for two typical CAD

```
uint32_t rf_get_chirp_time(uint8_t bw, uint8_t sf)
{
    const uint32_t bw_table[4]= {62500, 125000, 250000, 500000};

    if(bw < BW_62_5K || bw> BW_500K)
    {
        return 0;
    }

    return (1000000 / bw_table[bw - BW_62_5K]) * (1<< sf).
}
```

applications is given below:

1. The CAD function is used for channel detection before launching to ensure that the current channel is free, followed by data launching to avoid collision interference of wireless signals and improve the communication success rate.

CAD IO interrupt handler function that sets the CAD event to ACTIVE.

```
void Cad_Irq_Callback(void)
{
    rf_set_cad_detect_flag(MAC_EVT_TX_CAD_ACTIVE).
}
```

CAD timer timeout callback function that sets the CAD event to TIMEOUT.

```
void cad_tx_timeout_cb(void)
{
    stimer_stop(&stimer_txcad_event); rf_set_cad_detect_flag(MAC_EVT_TX_CAD_TIMEOUT);
}
```

CAD detection function, configures to enter the CAD receive state, sets the CAD event to NONE, and turns on the timer.

```
uint32_t rf_cad_detect_start(void)
{
        uint32_t bw, sf.
        uint32_t one_chirp_time; bw =
        rf_get_bw();
        sf= rf_get_sf();
        one_chirp_time = rf_get_chirp_time(bw, sf); // us RF_ASSERT(rf_cad_on(CAD_DETECT_THRESHOLD_10,
CAD_DETECT_NUMBER_3)).
        cad_tx_detect_flag= MAC_EVT_TX_CAD_NONE;
        RF_ASSERT(rf_enter_continous_rx()) SET_TIMER_MS(one_chirp_time *
        7 / 1000 + 1); return OK;
}
```

Wait and detect the CAD event. If it is TIMEOUT, then you can transmit immediately, if it is ACTIVE, then you need to back off and wait for some time before transmitting, and then you need to check the channel status again as described above before transmitting again.

2. The CAD function is used for channel detection before reception, which is used to check whether there is a useful signal in the current channel, and then decide, either to continue reception, or to turn off reception and go into standby or hibernation state to reduce power consumption.

```
bool check_cad_rx_inactive(uint32_t one_chirp_time)
{
        rf_delay_us(one_chirp_time * 7).
        rf_delay_us(360); // state machine start up time after enter rx state

        if (CHECK_CAD() ! = 1)
        {
                rf_set_mode(RF_MODE_STB3); return
                LEVEL_INACTIVE.
        }
```

```
        return LEVEL_ACTIVE.
}
```

Judge the CAD detection result according to the return value, if it is LEVEL_ACTIVE, then continue to receive and wait for the reception result; if it is LEVEL_INACTIVE, you can turn off the RX immediately to reduce power consumption.

```
// The main function can be called check_cad_rx_inactive using the following logic
while (1)
{
        if(check_cad_rx_inactive(one_chirp_time)== LEVEL_ACTIVE)
        {// Signal detected, waiting for reception to complete
         }else{
         //No signal detected, no need to continue to receive, can be dormant processing
         }
}
```

Examples of software applications for check_cad_rx_inactive and rf_cad_detect_start can be found in the CAD routines.